

# VISITING CARD SCANNER

## Developer Requirement and Implementation Planning Document

Web-based application now; Android app and SaaS commercialization in later phases

| Item                 | Details                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Name         | Visiting Card Scanner                                                                                                                                              |
| Prepared For         | Development Team                                                                                                                                                   |
| Primary Objective    | Scan or upload business/visiting cards, extract structured contact data, enrich company information from online sources, and categorize leads/cards for reporting. |
| Initial Release Type | Open-source web application without login.                                                                                                                         |
| Future Direction     | SaaS platform with user accounts, subscriptions, team workspaces, Android app, CRM integrations, and monetization.                                                 |
| Document Version     | v1.0                                                                                                                                                               |

### Executive Summary

The Visiting Card Scanner application will allow users to capture business card details either through a device camera or by uploading an image/PDF. The system will extract information such as person name, designation, phone number, email, company name, website, address, and other available fields. It will then optionally enrich the extracted company information using web search/company lookup logic and assign an appropriate business category such as Electrical Wires Supplier, Corporate Gifts Vendor, Packaging Supplier, IT Services, Real Estate, or other defined categories. The initial product will be a public web tool with no login, but the architecture should be designed so it can evolve into a SaaS product and later support an Android app using the same backend APIs.

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## 1. Product Vision and Objectives

The goal is to build a simple but professional web application that converts visiting/business cards into structured lead/contact records. The application should reduce manual data entry, improve lead organization, and help users categorize contacts based on company/business activity.

| Objective                 | Description                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| Fast capture              | Allow users to scan a physical visiting card using camera or upload an image/PDF.                                    |
| Accurate extraction       | Extract name, phone, email, company, designation, address, website, and other visible data.                          |
| Company enrichment        | Use available company name, website, email domain, and address to infer business type/category.                      |
| Simple public use         | Phase 1 should work without login and without complex onboarding.                                                    |
| Future-ready architecture | Design APIs, database, and modules so login, SaaS, Android app, subscriptions, and team features can be added later. |
| Export and reporting      | Allow users/admins to view, filter, download, and analyze extracted cards.                                           |

## 2. Scope: Phase 1, Phase 2, Future SaaS

| Phase                             | Scope                                                                                                                                              | Important Notes                                                       |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Phase 1 - Web MVP                 | Public web app. Camera scan. Image upload. PDF upload. OCR extraction. Manual edit before save/export. Basic category suggestion. CSV/XLSX export. | No mandatory login. Data retention should be configurable.            |
| Phase 2 - Improved AI and Android | Android app using same backend APIs. Better OCR, batch scan, image cropping, duplicate detection, lead notes, improved categories.                 | Backend should be API-first from Phase 1.                             |
| Phase 3 - SaaS                    | Login, teams, workspaces, subscriptions, usage limits, CRM integrations, admin dashboard, saved contacts, API keys.                                | Database should already support user_id/workspace_id nullable fields. |

## 3. User Types and Permissions

| User Type          | Phase            | Description                                      | Permissions                                                                       |
|--------------------|------------------|--------------------------------------------------|-----------------------------------------------------------------------------------|
| Guest User         | Phase 1          | Any public visitor using the tool without login. | Scan/upload, review extracted data, download output, optionally save temporarily. |
| Admin/Operator     | Phase 1 optional | Application owner or internal manager.           | View usage logs, review failed scans, maintain category list, manage settings.    |
| Registered User    | SaaS             | User with account/login.                         | Save cards, manage contacts, export data, use history, configure preferences.     |
| Team Admin         | SaaS             | Company account owner.                           | Manage users, billing, workspace settings, exports, permissions.                  |
| Developer/API User | SaaS optional    | User integrating scanner into another system.    | Use API keys, scan via API, retrieve structured JSON.                             |

## 4. Core User Flow

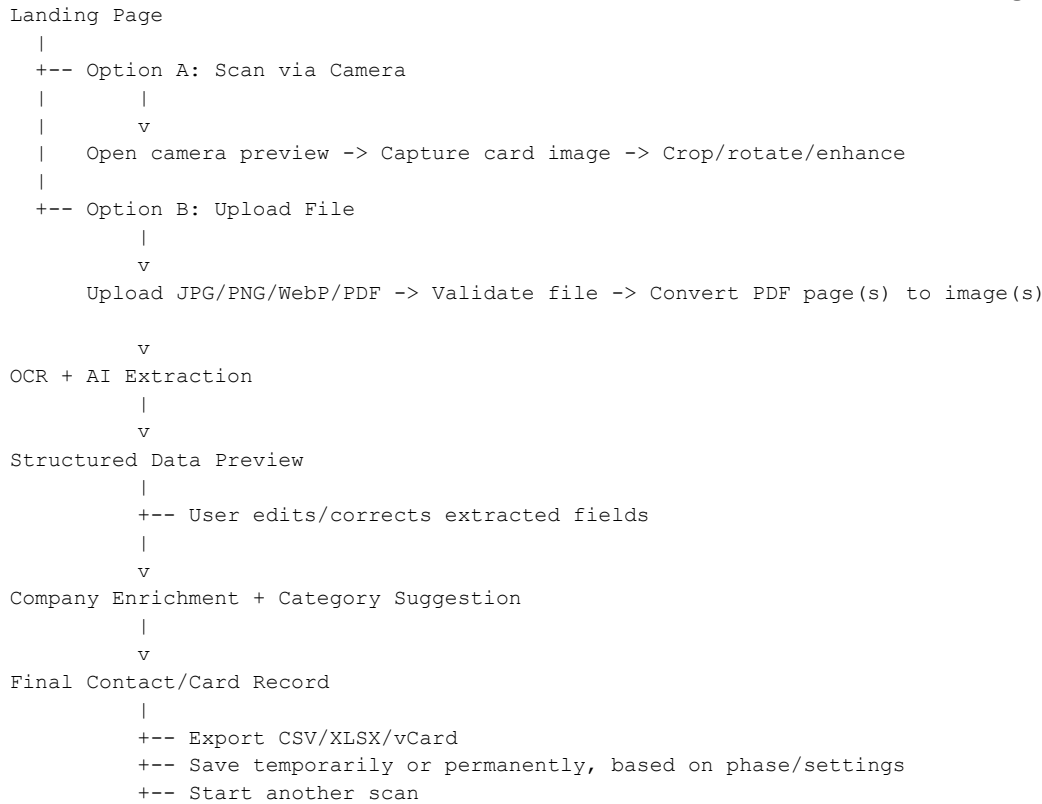
The application should keep the interaction simple: capture or upload, process, review, categorize, save/export.

USER JOURNEY FLOW

```

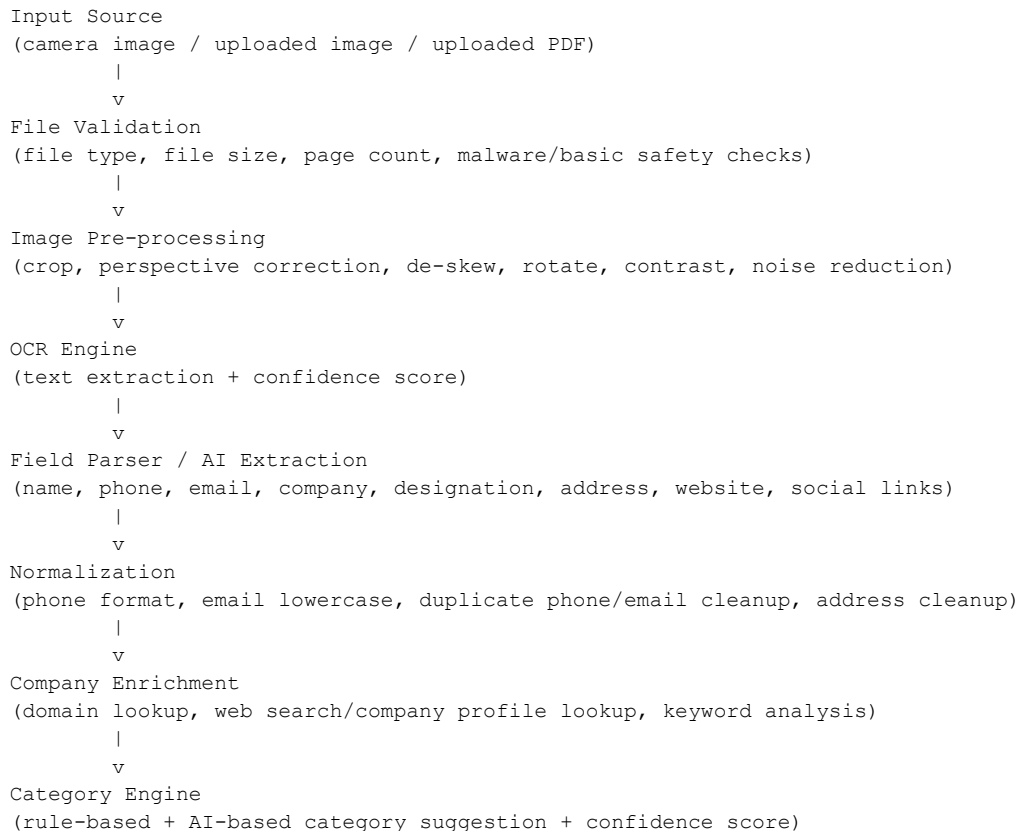
Start
 |
 v

```



## 5. Data Flow and Processing Pipeline

### DATA FLOW



```

    |
    v
Review UI
(user correction and approval)
    |
    v
Storage / Export / Report
(database, CSV, XLSX, vCard, API response)

```

| Pipeline Step    | Requirement                                                                        | Output                                  |
|------------------|------------------------------------------------------------------------------------|-----------------------------------------|
| Input capture    | Accept camera capture and upload. Supported uploads: JPG, JPEG, PNG, WebP, PDF.    | Raw file or captured image.             |
| Validation       | Reject unsupported, oversized, corrupted, password-protected, or unreadable files. | Validated input.                        |
| Pre-processing   | Crop, de-skew, rotate, sharpen, grayscale/contrast adjustment where useful.        | OCR-ready image.                        |
| OCR              | Extract raw text with line-level confidence if possible.                           | Raw OCR text.                           |
| Field extraction | Convert raw text into structured fields.                                           | JSON contact object.                    |
| Enrichment       | Use company/domain/address to infer business info and category.                    | Enriched metadata.                      |
| User review      | Allow correction before final output.                                              | Verified data.                          |
| Storage/export   | Save/export in useful formats.                                                     | Database record and downloadable files. |

## 6. OCR and Data Extraction Requirements

The OCR module should be separated from the application logic so the OCR provider can be changed later. For MVP, the developer can use an open-source OCR engine, but the system should support a paid cloud OCR/AI OCR provider later if higher accuracy is required.

| Field          | Extraction Rule / Notes                                                                       |
|----------------|-----------------------------------------------------------------------------------------------|
| Person Name    | Detect likely human name from prominent card text. Support Indian and international names.    |
| Designation    | Detect role titles such as Director, Sales Manager, Proprietor, Founder, CEO, Marketing Head. |
| Company Name   | Detect organization/business name from logo/title lines.                                      |
| Mobile / Phone | Extract mobile, landline, WhatsApp, telephone, fax where available. Normalize country code.   |
| Email ID       | Extract valid email addresses. Lowercase and validate format.                                 |
| Website        | Extract website/domain. If website missing, derive possible domain from email.                |
| Address        | Extract full address including city, state, PIN/ZIP, country where available.                 |
| Social Links   | Extract LinkedIn, Instagram, Facebook, YouTube, QR code URLs if present.                      |
| GST / Tax ID   | Optional field for Indian business cards if GSTIN or tax details appear.                      |
| Other Notes    | Store remaining useful card text that could not be classified.                                |

### 6.1 Extraction JSON Format

```

{
  "person_name": "",
  "designation": "",
  "company_name": "",
  "phones": [
    { "type": "mobile", "value": "", "country_code": "+91" }
  ],
  "emails": [""],
  "website": "",
  "address": {
    "full_address": "",
    "city": "",
    "state": "",
    "postal_code": ""
  }
}

```

```
    "country": ""
  },
  "social_links": [],
  "gst_in_or_tax_id": "",
  "raw_ocr_text": "",
  "extraction_confidence": 0.0,
  "category": {
    "suggested_category": "",
    "confidence": 0.0,
    "reason": ""
  },
  "enrichment": {
    "source": "",
    "company_keywords": [],
    "company_summary": ""
  }
}
```

7. Company Enrichment and Category Assignment

The system should not rely only on the text printed on the card. If company name, website, or email domain is available, the backend should attempt to infer the company activity and suggest a category. In Phase 1, use a hybrid approach: rules first, then AI/web enrichment if enabled.

| Input Signal   | How to Use It                                                                                                       |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Company name   | Search or match against known keywords. Example: "ABC Cables" may suggest wires/cables/electrical category.         |
| Website/domain | Fetch homepage meta title/description where allowed, or use search snippets.                                        |
| Email domain   | If email is name@company.com, derive company.com and use it for enrichment.                                         |
| Address/city   | Useful for regional reports and duplicate detection.                                                                |
| Card keywords  | Use product/service words printed on card: cables, wires, packaging, interiors, logistics, software, printing, etc. |

7.1 Category Assignment Logic

CATEGORY ENGINE LOGIC

- 1. Extract keywords from OCR text, company name, website text, and enrichment result.
- 2. Normalize words: lowercase, remove punctuation, use synonyms.
- 3. Match against category keyword dictionary.
- 4. If multiple categories match, rank by:
  - keyword match count
  - keyword importance
  - source reliability: website > card text > search snippet > guessed domain
  - AI confidence score, if AI model is used
- 5. Assign category only if confidence is above configured threshold.
- 6. If confidence is low, mark as "Uncategorized" and ask user/admin to select category manually.
- 7. Store category reason for transparency.

| Category Output    | Description                                                               |
|--------------------|---------------------------------------------------------------------------|
| suggested_category | The category name selected by the system.                                 |
| confidence         | Numeric score from 0 to 1.                                                |
| reason             | Human-readable reason, e.g., "Matched keywords: cable, wire, electrical". |
| needs_review       | True when confidence is low or multiple categories conflict.              |

## 8. Front-end Recommendations

The front-end should be professional, fast, responsive, and not unnecessarily bulky. It should work well on desktop browsers, mobile browsers, and later share design patterns with the Android app.

| Option                    | Recommended Use                                                      | Pros                                                                        | Cons                                          |
|---------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------|
| React + Vite + TypeScript | Best MVP choice if the developer wants a modern SPA.                 | Fast, lightweight, strong ecosystem, good camera/file handling support.     | SEO/server rendering not built-in by default. |
| Next.js + TypeScript      | Best if SEO, landing pages, auth, and SaaS growth are expected soon. | Professional-grade, scalable, good routing, API integration, SaaS-friendly. | Slightly heavier than Vite for a simple MVP.  |
| Vue 3 + Vite + TypeScript | Good alternative if developer prefers Vue.                           | Clean, lightweight, easy to develop.                                        | Smaller enterprise ecosystem than React.      |
| SvelteKit                 | Good for highly lightweight UI and fast performance.                 | Very clean and fast.                                                        | Developer availability may be lower.          |

Recommended front-end stack for this project: React + Vite + TypeScript for Phase 1 MVP, or Next.js + TypeScript if the team wants to directly prepare for SaaS, SEO, login, pricing pages, and server-side workflows. Use Tailwind CSS or a clean component library such as shadcn/ui-style components for professional UI without heavy styling overhead.

| UI Area              | Requirement                                                                |
|----------------------|----------------------------------------------------------------------------|
| Landing page         | Explain scan/upload options, supported formats, privacy note, and CTA.     |
| Scanner screen       | Camera preview, capture button, retake, crop/rotate, continue.             |
| Upload screen        | Drag-and-drop upload, supported file notice, progress indicator.           |
| Processing screen    | Show upload/OCR/enrichment progress with clear status messages.            |
| Review screen        | Editable extracted fields, confidence indicators, raw text toggle.         |
| Category screen      | Show suggested category, confidence, reason, and manual category dropdown. |
| Result/export screen | Download CSV/XLSX/vCard, copy JSON, start another scan.                    |

## 9. Backend and Architecture Recommendations

The backend should be API-first. The same APIs should be usable by the web app now and Android app later. Keep OCR, extraction, enrichment, category assignment, and export as separate modules/services.

| Layer              | Recommended Responsibility                                                                             |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Frontend           | Camera capture, file upload, field review, export actions.                                             |
| API Backend        | Authentication later, upload handling, OCR orchestration, data normalization, category logic, storage. |
| OCR Service        | Image/PDF text extraction and OCR confidence scoring.                                                  |
| AI/Parser Service  | Convert OCR text into structured contact fields.                                                       |
| Enrichment Service | Company/domain lookup, website metadata, search snippet analysis where allowed.                        |
| Database           | Store scans, contacts, categories, exports, users/workspaces in future.                                |
| Queue/Worker       | Handle OCR/enrichment asynchronously for larger files or batch upload.                                 |

| Backend Option           | Recommended Use                                      | Notes                                                              |
|--------------------------|------------------------------------------------------|--------------------------------------------------------------------|
| Node.js + NestJS/Fastify | Good if developer is JavaScript/TypeScript oriented. | Professional architecture, API-first, scalable.                    |
| Python + FastAPI         | Strong choice for OCR, AI, PDF/image processing.     | Excellent for ML/OCR workflows and clean APIs.                     |
| PHP/Laravel              | Good if the existing team/server stack is PHP-based. | SaaS features are mature, but OCR/AI workers may still use Python. |

Recommended backend for OCR-heavy implementation: Python FastAPI for OCR/extraction services, or a hybrid where the main app is Node/PHP and OCR/AI processing runs as a Python microservice.

## 10. Database Design

Even though Phase 1 may not require login, the database should include nullable fields for future SaaS support. This avoids major migration later.

| Table              | Purpose                                          | Key Fields                                                                                                          |
|--------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| scans              | One uploaded/scanned file processing job.        | id, user_id nullable, workspace_id nullable, input_type, file_path, status, created_at, processed_at, error_message |
| contacts           | Structured extracted visiting card/contact data. | id, scan_id, person_name, designation, company_name, website, address_json, raw_text, confidence, created_at        |
| contact_phones     | Multiple phone numbers per contact.              | id, contact_id, type, number, country_code                                                                          |
| contact_emails     | Multiple emails per contact.                     | id, contact_id, email                                                                                               |
| categories         | Business category master.                        | id, name, parent_id, keywords_json, is_active                                                                       |
| contact_categories | Final/suggested category mapping.                | id, contact_id, category_id, confidence, reason, source, is_user_confirmed                                          |
| enrichment_logs    | Company enrichment attempts and sources.         | id, contact_id, source_type, source_url, summary, keywords_json, created_at                                         |
| exports            | Export history.                                  | id, user_id nullable, format, file_path, record_count, created_at                                                   |
| users              | Future SaaS users.                               | id, name, email, password_hash, plan_id, status, created_at                                                         |
| workspaces         | Future company/team accounts.                    | id, name, owner_user_id, plan_id, created_at                                                                        |

## 11. Screens and UI Requirements

| Screen                | Main Elements                                                                             | Developer Notes                          |
|-----------------------|-------------------------------------------------------------------------------------------|------------------------------------------|
| Home/Landing          | Title, description, Scan via Camera button, Upload File button, privacy note.             | Keep simple and fast.                    |
| Camera Scanner        | Camera permission, preview, capture, retake, crop/rotate.                                 | Handle no camera gracefully.             |
| File Upload           | Drag-drop upload, file picker, supported formats, file size limit.                        | Support PDF/image.                       |
| Processing            | Progress steps: Uploading, OCR, extracting fields, enriching company, assigning category. | Use clear loading states.                |
| Review Extracted Data | Editable form fields with confidence indicators.                                          | User must be able to correct all fields. |
| Category Confirmation | Suggested category, confidence, reason, manual dropdown, add custom category if allowed.  | Avoid black-box category decisions.      |
| Results/Export        | Final card preview, export buttons, scan another card.                                    | CSV and XLSX minimum. vCard optional.    |
| Admin Settings        | Category master, keyword mapping, logs.                                                   | Optional for Phase 1 but useful.         |

## 12. API Requirements

| Endpoint                | Method | Purpose                                        | Request/Response Notes                     |
|-------------------------|--------|------------------------------------------------|--------------------------------------------|
| /api/scans              | POST   | Upload image/PDF or captured image.            | Return scan_id and status.                 |
| /api/scans/{id}/status  | GET    | Check processing status.                       | Return queued/processing/completed/failed. |
| /api/scans/{id}/result  | GET    | Get OCR and structured extraction result.      | Return JSON contact object.                |
| /api/contacts           | POST   | Save reviewed contact.                         | Accept corrected fields and category.      |
| /api/categories         | GET    | List active categories.                        | Used in category dropdown.                 |
| /api/categories/suggest | POST   | Suggest category from extracted/enriched data. | Return ranked category list.               |
| /api/exports            | POST   | Export selected contacts.                      | CSV/XLSX/vCard.                            |
| /api/health             | GET    | Health check.                                  | Useful for deployment monitoring.          |



## 13. Reporting and Export Requirements

| Report / Export     | Requirement                                                     |
|---------------------|-----------------------------------------------------------------|
| CSV Export          | Export contact fields, category, confidence, created date.      |
| XLSX Export         | Formatted spreadsheet with columns and filters.                 |
| vCard Export        | Optional export to phone/contact managers.                      |
| Category Report     | Count of cards by category.                                     |
| City/Region Report  | Count by city/state where address can be parsed.                |
| Scan Quality Report | Failed scans, low-confidence extraction, fields missing.        |
| Duplicate Report    | Duplicate contacts by email, phone, company/person combination. |

## 14. Security, Privacy, and Compliance

- Display a clear privacy note: cards may contain personal/business contact data.
- For open public Phase 1, define whether uploaded files are stored permanently, temporarily, or deleted after processing.
- Apply file size limits and supported file type validation.
- Do not expose uploaded file URLs publicly without signed/temporary access.
- Sanitize OCR text and user input before storing or rendering.
- Add rate limits to prevent abuse, especially since there is initially no login.
- Keep API keys for OCR/AI/search providers only on the backend.
- Maintain audit logs when the SaaS version is introduced.
- Provide a delete option for stored scans/contacts in SaaS version.

## 15. Error Handling and Data Quality

| Scenario                 | Expected Handling                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------|
| No camera available      | Show upload option and message: "Camera not available on this device."                              |
| Camera permission denied | Explain how to allow camera or use upload instead.                                                  |
| Poor image quality       | Ask user to retake or upload a clearer image. Show tips: good light, full card visible, avoid blur. |
| OCR fails                | Show raw error, allow retry, store failure reason for admin.                                        |
| PDF has multiple pages   | Allow selecting page or process all pages as separate cards.                                        |
| Missing company name     | Do not run company enrichment; category remains Uncategorized or based on card keywords.            |
| Category confidence low  | Show manual category dropdown.                                                                      |
| Duplicate contact        | Warn user and allow merge/keep both.                                                                |

## 16. Suggested Categories

The category list should be configurable. Below is a starting master list. The developer should implement category keyword mapping so admins can add/edit categories without code changes.

| Category                       | Example Keywords                                                      |
|--------------------------------|-----------------------------------------------------------------------|
| Electrical Wires Supplier      | wires, cables, electricals, copper, switches, MCB, LED, lighting      |
| Corporate Gifts Supplier       | gifts, promotional products, branding, trophies, hampers, merchandise |
| Printing and Packaging         | printing, box, packaging, offset, digital print, label, carton        |
| IT Services and Software       | software, website, app, ERP, CRM, cloud, hosting, IT services         |
| Digital Marketing Agency       | SEO, social media, ads, branding, marketing, performance marketing    |
| Real Estate and Construction   | builders, developers, construction, architect, interior, contractor   |
| Logistics and Transport        | logistics, courier, transport, freight, cargo, warehouse              |
| Manufacturing                  | manufacturer, factory, production, components, OEM, machinery         |
| Wholesale Trader / Distributor | dealer, distributor, wholesaler, trading, stockist                    |
| Finance and Accounting         | CA, accountant, tax, GST, finance, loan, insurance                    |
| Healthcare and Medical         | clinic, hospital, pharma, medical, doctor, diagnostics                |
| Education and Training         | school, institute, academy, coaching, training                        |
| Food and Hospitality           | restaurant, catering, hotel, bakery, food products                    |

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| Textile and Garments | textile, garments, fabric, apparel, clothing, uniform |
| Security Services    | security, CCTV, guard, surveillance, access control   |
| Uncategorized        | Fallback when confidence is low.                      |

## 17. Roadmap and Milestones

| Milestone                        | Deliverables                                                                    | Acceptance Output                                     |
|----------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------|
| M1 - Project Setup               | Repo, backend skeleton, frontend skeleton, database schema, environment config. | App runs locally with health check.                   |
| M2 - Upload and Camera Capture   | Camera capture, image upload, PDF upload, validation.                           | User can submit card input.                           |
| M3 - OCR Pipeline                | Image preprocessing, OCR extraction, raw text view.                             | Raw card text extracted.                              |
| M4 - Structured Field Extraction | Parser/AI extraction to JSON, editable review form.                             | Name, phone, email, company, address fields populate. |
| M5 - Category Engine             | Category master, keyword mapping, category suggestion with confidence.          | Suggested category appears with reason.               |
| M6 - Export and Reporting        | CSV/XLSX export, basic dashboard/report.                                        | User can download structured data.                    |
| M7 - Admin Settings              | Category management and logs.                                                   | Admin can manage categories and review failures.      |
| M8 - Android/API Readiness       | Documented APIs, token support placeholder, mobile-friendly upload.             | Same backend can support Android app.                 |

## 18. Acceptance Criteria

- User can scan a card using a camera from supported browsers/devices.
- User can upload image files and PDF files.
- Application extracts at least name, phone, email, company name, and address when visible on the card.
- User can review and manually edit every extracted field before final save/export.
- Application suggests a category with confidence score and reason.
- If category confidence is low, system marks the card as Uncategorized or asks user to choose category.
- User can export at least CSV and XLSX.
- System handles missing camera, poor image, unsupported file, OCR failure, and multi-page PDF gracefully.
- Backend APIs are documented and reusable by a future Android app.
- Code is modular: OCR, parser, enrichment, category engine, export, and storage should be separate modules.

## 19. Developer Notes and Open Decisions

| Decision Area       | Recommendation / Question                                                                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OCR engine          | Start with open-source OCR for MVP, but keep provider abstraction so cloud OCR can be added.                                                                     |
| AI parser           | Use deterministic regex for phone/email/website and AI/heuristics for name/company/designation/category.                                                         |
| Temporary storage   | Decide whether public scans are deleted after export or retained for analytics.                                                                                  |
| Login               | Do not build mandatory login in Phase 1, but keep database fields ready for user/workspace.                                                                      |
| SaaS billing        | Future: plans based on scans/month, batch scans, exports, team seats, API access.                                                                                |
| Category ownership  | Allow global categories initially; SaaS version may allow workspace-specific custom categories.                                                                  |
| Internet enrichment | Define whether enrichment uses website scraping, search API, AI search, or manual keyword dictionary. Must respect provider terms and robots/policy constraints. |
| Android app         | Build later using the same API endpoints; avoid web-only backend assumptions.                                                                                    |

## Appendix A: Core User Stories

| User Story                                                                                                     | Acceptance                                          |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| As a guest user, I want to scan a visiting card using my camera so that I can extract contact details quickly. | Camera opens, capture works, image is processed.    |
| As a guest user, I want to upload a PDF or image so that I can process cards saved on my device.               | Upload validates and processes supported files.     |
| As a user, I want to correct extracted information before export so that my final data is accurate.            | Editable review form is available.                  |
| As a user, I want the app to suggest a business category so that I can organize contacts automatically.        | Category suggestion appears with confidence/reason. |
| As an admin, I want to manage categories and keywords so that category reports improve over time.              | Category management screen/API exists.              |
| As a future SaaS user, I want saved scan history so that I can manage leads over time.                         | Architecture supports user/workspace ownership.     |